Kill Switch Module — Experiments

1. Verification of MazF Killing Activity

MazF is the toxin of the classical mazEF toxin–antitoxin system; it has mRNA endoribonuclease activity and serves as the core lethal element in the suicide-switch design of this project. To verify the lethality and killing efficiency of MazF in the eukaryotic chassis *Dunaliella salina*, we constructed the verification plasmid pCAMBIA1302-CaMV 35S-mazF, in which *mazF* is driven by the strong constitutive CaMV 35S promoter, and performed functional characterisation.

1) Plasmid construction: the *mazF* gene was cloned downstream of the CaMV 35S promoter in the pCAMBIA1302 backbone by restriction-enzyme digestion, generating p1302-CaMV 35S-mazF. The plasmid was transformed into *E. coli* DH5α competent cells for cloning and amplification; single colonies were picked for colony PCR screening. Clear specific bands appeared at the expected size on an agarose gel, and sequencing confirmed that *mazF* was correctly inserted into the backbone.

2) Osmotic transformation: p1302-CaMV35S-mazF was delivered into *Dunaliella salina* by osmotic transformation. Log-phase algal cells were collected, pre-adapted in 1 M NaCl medium for 8 h, resuspended in 0.1 M NaCl medium, and mixed successively with 5.46 μg plasmid DNA and 15 μL of 0.1% *Triton X-100*. After vortexing for 90 s, cells were allowed to recover for 48 h.

3) Hygromycin selection: the vector carries the *hptII* hygromycin-resistance marker. After transformation, algal cells were plated on hygromycin B-containing medium to obtain positive transformants.

4) Growth-curve assay: untransformed wild type (WT) and empty-vector transformants (EV) served as negative controls. The three groups were inoculated at the same starting cell density in PKS/DM liquid medium containing 0.5 M NaCl (26 °C, 50 μmol photons m^{-2} s^{-1}, 16 h light / 8 h dark photoperiod), and OD_{680} was measured daily for 10 consecutive days to plot growth curves.

2. Validation of the P_{DCA1} Salt-Induced Response System

P_{DCA1} is an endogenous salt-regulated promoter of *Dunaliella salina*: its transcriptional activity is significantly enhanced under high-salt conditions and weakened under low-salt conditions. Verifying its salt-responsive characteristics is a prerequisite for designing the "survive in high salt, die in low salt" control logic of the suicide switch.

1) Plasmid construction: the endogenous P_{DCA1} promoter and the *EGFP* reporter gene were assembled into the linearised pCAMBIA2301 backbone by homologous recombination, generating p2301-P_{DCA1}-EGFP. The assembly product was transformed into *E. coli* DH5α competent cells; single colonies were picked for colony PCR screening, which preliminarily confirmed insertion of the target fragment. Sequencing confirmed that all functional elements were correctly assembled.

2) Salt-gradient treatment: transformants were inoculated and cultured in media containing different NaCl concentrations (0.5 M, 2 M and 3 M).

3) Reporter assay: EGFP fluorescence was measured to quantify P_{DCA1} transcriptional activity as a function of salinity and to determine its salt-responsive dynamic range.

3. Co-transformation and Functional Validation of the Switch Module

Strong-promoter (CaMV 35S)-driven toxin expression is too high for cells to neutralise, making transformants difficult to maintain stably. The final circuit therefore needs to down-regulate toxin expression to a moderate level: the pCAMBIA1302-Bar backbone, in which the phosphinothricin-resistance marker (Bar) is substituted, is used, and the P_{ip} promoter—an endogenous promoter of low expression abundance in *Dunaliella salina*—is chosen to drive *mazF*. In parallel, the p1302-P_{DCA1}-*mazE* vector is constructed, in which P_{DCA1} drives the antitoxin gene *mazE*. The two plasmids are co-transformed into *Dunaliella salina*; transformants carrying both the toxin and antitoxin modules are obtained by orthogonal double-resistance selection, and the complete "survive in high salt, die in low salt" switching function is validated.

1) Toxin vector preparation: the P_{ip} promoter and *mazF* gene fragment were amplified by PCR; the pCAMBIA1302-Bar backbone was linearised, and p1302-Bar-P_{ip}-mazF was assembled by multi-fragment homologous recombination. The product was transformed into *E. coli* DH5α and plated on kanamycin plates (50 and 100 μg/mL) for selection. Single colonies were picked for colony PCR verification; electrophoresis showed single bright target bands in lanes 1, 2, 3, 6 and 8, with no specific amplification in the negative control. Sequencing confirmed the correct sequence.

2) Antitoxin vector preparation: using pCAMBIA2301 as the backbone, the P_{DCA1} promoter and the *mazE* antitoxin gene were assembled by homologous recombination to generate p2301-P_{DCA1}-mazE. After transformation into *E. coli* DH5α, cells were plated on kanamycin plates for selection; positive clones were identified by colony PCR, and sequencing confirmed correct assembly of the functional elements.

3) Co-transformation: p1302-Bar-P_{ip}-mazF and p2301-P_{DCA1}-mazE were mixed at an equal molar ratio and simultaneously delivered into *Dunaliella salina* by osmotic transformation: 1 M NaCl pre-adaptation for 8 h, resuspension in 0.1 M NaCl, addition of the mixed plasmid DNA and 0.1% Triton X-100, vortexing for 90 s, and recovery culture for 48 h.

4) Double-resistance selection: after transformation, algal cells were plated on double-selective PKS/DM medium containing phosphinothricin (selecting for p1302-Bar-P_{ip}-mazF) and G418 (selecting for p2301-P_{DCA1}-mazE), to obtain double-positive transformants carrying both plasmids.

5) Salt-switching functional assay: double-positive transformants were inoculated in parallel into high-salt (2 M NaCl) and low-salt (0.5 M NaCl) media, and OD_{680} was measured continuously. The switching phenotype was observed: under high salt, P_{DCA1} is activated and drives MazE expression, which neutralises MazF toxicity and allows cells to survive; under low salt, P_{DCA1} activity weakens, MazE expression becomes insufficient, MazF becomes relatively excessive, and cells die.